

KUUSAMO RUKATUNTURI, Finland

WMO: 028680

Lat: 66.1661N

Lon: 29.1517E

Elev: 1190

StdP: 14.07

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-3.9	-12.7	2.9	-8.0	-8.1	3.7	-3.5	42.1	15.9	38.8	16.5	20.9	100	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	73.6	60.5	69.7	58.5	66.4	56.9	62.8	69.2	60.7	66.8	58.7	64.6	12.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.5	82.2	64.8	58.1	75.4	62.9	56.1	70.0	61.0	29.0	69.4	27.4	67.2	26.0	64.8	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.5	31.9	29.2	-10.9	75.8	6.4	4.3	-15.5	78.9	-19.3	81.4	-22.9	83.9	-27.6	87.0
			-10.2	64.8	6.5	3.0	-14.9	67.0	-18.6	68.8	-22.3	70.4	-27.0	72.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.7	11.5	14.2	19.6	29.3	40.9	49.6	56.8	52.4	43.8	31.1	23.4	18.0	(6)
(7)		DBStd	16.88	8.81	9.36	7.06	6.69	9.19	8.25	7.30	6.68	6.10	6.17	6.64	7.95	(7)
(8)		HDD50	6825	1194	1002	944	620	312	106	22	50	198	586	798	993	(8)
(9)		HDD65	11830	1659	1422	1409	1070	749	467	268	395	635	1050	1248	1458	(9)
(10)		CDD50	493	0	0	0	0	29	93	233	125	13	0	0	0	(10)
(11)		CDD65	23	0	0	0	0	0	5	13	5	0	0	0	0	(11)
(12)		CDH74	68	0	0	0	0	1	20	32	15	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.7	18.9	18.9	18.2	16.7	16.2	14.5	13.5	14.2	15.3	16.5	18.4	18.7	(14)
(15)	Precipitation	PrecAvg	22.4	1.5	1.2	1.2	1.1	1.8	2.5	3.2	2.5	2.0	1.9	1.7	1.6	(15)
(16)		PrecMax	28.8	2.9	3.0	2.0	1.7	3.1	5.0	5.6	5.5	3.8	3.5	3.9	2.9	(16)
(17)		PrecMin	17.5	0.7	0.2	0.4	0.4	0.7	0.7	1.2	1.0	0.4	0.3	0.3	0.4	(17)
(18)		PrecStd	3.1	0.6	0.7	0.5	0.4	0.6	1.1	1.0	1.0	0.9	0.9	0.8	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	32.4	37.2	52.2	70.2	76.6	77.5	76.3	61.6	47.4	39.3	31.8	(19)
MCWB			31.0	32.3	33.4	42.8	52.8	60.7	62.8	64.2	54.4	47.2	37.6	31.8	(20)	
2%		DB	28.3	30.8	33.0	45.6	66.0	71.9	74.3	69.3	57.3	43.2	36.0	30.8	(21)	
		MCWB	28.1	30.7	30.9	38.4	52.4	58.5	61.3	59.8	52.8	42.6	35.3	30.7	(22)	
5%		DB	26.1	29.0	30.9	40.9	60.6	66.0	71.0	65.4	54.3	41.3	33.6	29.5	(23)	
		MCWB	25.8	28.8	29.6	35.5	50.0	55.4	59.2	57.8	50.5	40.8	33.5	29.4	(24)	
10%		DB	23.5	26.7	29.0	38.2	54.8	62.0	67.8	61.8	51.9	39.3	31.7	28.1	(25)	
		MCWB	23.1	26.5	28.2	33.9	46.2	52.5	58.3	56.0	49.4	38.8	31.6	27.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.1	32.3	34.2	43.2	55.5	63.3	65.7	65.2	56.4	47.2	38.0	31.8	(27)
MCD B			31.2	32.4	36.3	49.2	63.8	70.5	72.0	73.8	59.3	47.5	38.0	31.8	(28)	
2%		WB	28.2	30.8	31.5	39.7	53.1	60.0	63.4	61.2	53.2	42.9	35.3	30.6	(29)	
		MCD B	28.2	30.8	32.4	44.7	63.9	68.7	69.6	66.4	55.8	43.0	35.4	30.6	(30)	
5%		WB	26.0	28.9	30.0	36.5	50.6	57.0	61.8	58.9	51.5	40.9	32.9	29.3	(31)	
		MCD B	26.2	29.0	30.8	39.8	60.2	64.9	67.9	63.5	53.5	41.1	33.0	29.4	(32)	
10%		WB	23.2	26.6	28.4	34.7	47.1	53.9	59.9	57.0	49.8	38.9	31.3	27.8	(33)	
		MCD B	23.4	26.8	28.9	37.2	54.4	59.5	66.0	60.9	51.4	39.2	31.4	27.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	7.9	8.6	8.7	11.5	11.6	10.0	7.7	5.6	6.1	6.4	(35)	
MCDBR			8.7	7.7	8.7	11.6	18.2	15.8	14.5	13.6	10.5	7.7	7.3	6.2	(36)	
MCWBR			9.2	7.7	7.8	7.2	10.1	8.1	7.3	7.4	7.0	7.1	6.9	6.5	(37)	
5% WB		MCD BR	8.7	7.7	8.1	10.2	17.6	15.1	13.4	11.9	9.5	7.6	6.2	6.4	(38)	
		MCWBR	9.1	7.8	7.6	6.8	10.3	9.1	8.1	7.8	7.0	7.3	6.4	6.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.189	0.256	0.289	0.321	0.334	0.334	0.350	0.330	0.307	0.278	0.350	0.190	(40)	
(41)		taud	1.923	2.112	2.170	2.250	2.460	2.496	2.467	2.498	2.546	2.366	2.731	1.947	(41)	
(42)		Ebn at Noon	76	190	243	266	276	279	269	264	246	188	81	10	(42)	
(43)		Edh at Noon	5	17	27	32	30	29	29	25	19	14	6	1	(43)	
(44)	All-Sky Solar Radiation	RadAvg	27	186	651	1214	1504	1605	1494	1087	582	192	43	4	(44)	
(45)		RadStd	2	35	61	117	140	117	151	108	60	32	7	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(46)											(46)
(47)	Regional (21 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
											(47)

Nomenclature: See separate page